"Causal drivers of climate-mediated coral reef regime shifts by Arif et al. (2022)": A Causal Inference Review

Jess Prevost

2026-10-16

Contents

1 Introduction **1**

2 Review **1**

2.1 Causal question and predictor variables 1

2.2 Directed Acyclic Graph (DAG) specification 2

2.3 Causal adjustment strategy 3

2.4 Bayesian analysis 4

2.5 Limitations of the study 4

3 Reproduction **5**

3.1 Causal analysis 5

3.2 DAG-informed predictive model 8

4 Conclusion **10**

References **11**

1 Introduction

This report sets out to critically review and attempt to replicate the analysis by Arif et al. (2022). This paper is the first causal analysis using observational data to study the potential causes of regime shifts of coral reefs following bleaching events. The analysis uses common causal inference frameworks introduced by Pearl (2009) including Directed Acyclic Graphs (DAGs) and the backdoor adjustment criterion.

2 Review

2.1 Causal question and predictor variables

The main causal question in this paper is well defined as: how did specific key variables each causally affect the recovery trajectories of coral reefs in the Seychelles following bleaching events in 1998? (Graham et

al. 2015). This is outlined in the abstract and revisited throughout the text. Rather than a single causal question, this analysis essentially examines a series of causal questions aimed at determining the effects of several variables on a single response variable. Causal inference is the prediction of intervention (Pearl 2009), and here the causal question is predicting whether changes in any of the key variables would change the recovery trajectory of a reef. The question is fairly specific indicating that the results apply only to these reefs in the Seychelles and this specific bleaching event. The authors recognise the limitations of the causal relationships by not attempting to generalise to other coral reef localities. The predictive model based off the causal analysis was however used to predict the recovery trajectory following a bleaching event in 2016 for the same original reefs which had since recovered from the 1998 bleaching event.

Some details about the predictor variables were omitted from the paper but these were mostly available in the earlier research by Graham et al. (2015). The alternative recovery trajectories of coral reefs were defined as a binary variable of recovering or regime shifting as per Graham et al. (2015). Recovering reefs had coral cover greater than macroalgal coverage and time series data showed high or increasing coral cover while regime shifting reefs showed macroalgae dominance greater than coral cover and macroalgal cover remained high or increased over time (Graham et al. 2015). Four metrics were used to define recovery trajectories with the full methodology described in Graham et al. (2015).

Data collection for the branching coral cover, macroalgal cover, herbivorous fish biomass, and structural complexity variables was stated as 1994 and could thus be considered as pre-bleaching conditions which is essential to ensure the direction of causation from these variables to the response variable. It may be a limitation however that the measurement of these variables was four years prior to the bleaching event. During this period, other unaccounted for factors may have caused changes in these variables, altering their effects on the response. Collection dates for the other variables prior were not stated but could be found in the literature (Graham et al. 2015). There was however a discrepancy relating the the data collection for the nutrient measurements. Arif et al. (2022) stated that these data were from 2004 and suggested Graham et al. (2015) as the source of the data but according to Graham et al. (2015) nutrient measurements were collected in 2014. This may have been a typographical error. The 2014 data from Graham et al. (2015) was unfortunately unavailable thus it could not be cross-checked. If this is not a typographical error, the source of the nutrient data should have been cited or data collection described. Either way, the nutrient measurements were taken after the bleaching event occurred. Nutrient content is expected to change over seasons but differences in sites are anticipated to remain stable (Graham et al. 2015). Nevertheless, there is the possibility for the post-bleaching macroalgal content to affect nutrient levels which would be an issue for causal inference from measurements taken post-bleaching. Depth, MPA status, and wave exposure data were expected to remain stable over time. Thus even though depth was measured and wave exposure was calculated post-bleaching (1998 - 2011), it should not be an issue for inference as macroalgae content post-bleaching should not affect depth, MPA status, or wave exposure.

2.2 Directed Acyclic Graph (DAG) specification

Directed Acyclic Graphs are a commonly utilised tool for specifying causal models (Pearl 2009). The specification of the DAG for the causal question here was appropriate as it used domain knowledge and extensive research on the issue of coral bleaching. There was a detailed explanation in the supplementary text for each of the causal relationship included in the DAG. I inspected the stated relationships and they were all justifiable. Despite this, other important variables may still be excluded from the DAG such as substrate type (Graham et al. 2015). There were also no potentially unobserved variables included in the DAG. It is possible that no other variables were recognised to be related to any of the variables in the current DAG. Alternatively, there is a risk that that the DAG was created around the variables for which data were available, especially since this analysis was done retrospectively on data previously collected for a "causal salad" model of coral reef bleaching. The "causal salad" model is where all predictor variables are placed in a single model with each of their coefficients often incorrectly interpreted as the total effect on the response (McElreath 2020). Preferably, the causal model should be defined first and used to guide data collection (Pearl 2009). Although this is tricky when data on the covariates from before the bleaching event is required to ensure it is the causal effect of the predictor on the response. If important variables were omitted, there

could be detrimental effects on inference from the causal models. For example, if a fork or a common cause of the response, whether a reef has shifted regimes or not, and one of the direct predictor variables, exists and this is not conditioned one, the estimate of the causal effect of the variable on the response will be biased (Pearl 2009).

There was discrepancy between the DAG downloaded from dagitty.net in the supplied code and the DAG stated in the paper (Arif et al. 2022). The downloaded DAG contained a directed edge from depth to initial macroalgae cover and excluded the directed edge from wave exposure to branching coral unlike the DAG in the paper. The causal relationships described and the d-separation conditions examined in the supplementary information were associated with the DAG in the paper. After checks it was found that the data were in fact consistent with the d-separation conditions implied by both DAGs. The d-separation method is a way to test for DAG-data consistency (Pearl 2009). In addition to a DAG being defensible based on ecological theory, all d-separation conditions or independencies implied by a given DAG should be consistent with the data in order for the DAG to be utilised. These inconsistencies in DAG structure does raise the question of whether the final causal model stated in the paper was the first to be proposed or whether alternative DAGs, such as the one in the supplied code, were tested. The authors recognise that multiple DAGs can pass DAG-data consistency, and that the final DAG should be justified with theory but do not outline their consideration of alternative DAGs. Ideally, an analysis of alternative DAGs and explanation of why the final causal model was chosen should have been included, especially since the data were consistent with both of these DAGs yet DAG choice may affect the causal models specifications and thus results.

2.3 Causal adjustment strategy

The authors used the backdoor criterion to define an adjustment set for each variable in order to estimate the total causal effect of the given variable on the recovery trajectory of coral reefs. This is an established method for calculating sets of variables to control for to give an unbiased estimate of the effect of one variable on another (Pearl 2009). The method involves blocking all backdoor paths entering the causal variable by conditioning on the relevant variables but ensuring the wrong variables, such as colliders are not adjusted for. All the models specified in the paper satisfy the backdoor criterion for the DAG used in the paper. The model specifications for some of the variables would change if the DAG in the supplied code were used instead (such as branching coral, which would no longer require wave exposure to be conditioned on). For the majority of variables including MPA, depth, wave exposure, nutrients, branching, structural complexity, and herbivore biomass, the minimal adjustment set was used for the model specifications, meaning the set containing the fewest variables which satisfies the backdoor criterion. However, for the initial macroalgae model, depth was included as a covariate even though the minimal adjustment set does not include this variable. The adjustment set still satisfies the backdoor criterion with depth included, it is just not the minimal adjustment set. The inclusion of depth might be as a result of the use of the DAG in the supplied code prior to the final DAG included in the paper. The supplementary text hints at this with the inclusion of paths 5 and 6 in section S1.3 which do not exist in the DAG in the paper but do exist in the DAG in the supplied code. The directed edge from depth to initial macroalgae opens additional backdoor paths which should be adjusted for by including depth when using the DAG in the supplied code. Reasons for the decision to use the minimal adjustment sets for the other variables and not to use the minimal adjustment set for the initial macroalgae model should have been explicitly stated. While all adjustment sets which satisfy the backdoor criterion give unbiased estimates of the total causal effect but they can differ in precision of the estimates depending on the additional variables included (Pearl 2009).

There were additional errors in the supplementary information for section S1.3. The initial macroalgae model had an incomplete path (path 4) listed, although this did not affect the adjustment set. The model specified in the paper for herbivore biomass which adjusts for structural complexity and MPA is the minimal set satisfying the backdoor criterion. However, the supplementary text description has an error, stating that only structural complexity and branching coral need to be adjusted for. This is incorrect and insufficient as an adjustment. Branching coral is a collider and opens another backdoor path (Herbivore biomass <- MPA -> Branching coral <- Depth -> Regime Shift) thus MPA or Depth would still additionally need to be included in the set to block this newly opened path.

2.4 Bayesian analysis

The paper used Bayesian statistical models to fit the causal models for each of the variables and the predictive model, all using the *rethinking* package (McElreath 2024). Bayesian models are useful as they yield full posterior distributions for the parameters of interest. The weakly informative priors for all the models were appropriate and it was stated that literature was consulted in selecting these weakly informative priors. A limitation in the implementation of the Bayesian models may be the use of only a single chain when running the models, thus not being able to properly assess convergence. There was no mention of convergence diagnostics in the paper at all which is important to check before assessing the parameter estimates. Additionally, there was no setting of the seed in the supplied code meaning the model fits were not completely reproducible.

The authors correctly only interpreted the coefficient of the predictor variable of interest as the causal effect on the response in each of the causal models (Arif et al. 2022). While the forest plot showing the standardised effect size and credible intervals for each of the predictor variables based on the posterior estimates from the Bayesian models is effective (Figure 3a), there is a loss of information compared to displaying the full posterior distribution. In this way, it may have been more appropriate to plot ridgeline plots for the effect sizes with the full posterior distributions. Ridgeline plots were used effectively for the probability of regime shift for the predictive model. The effect sizes are displayed on the log-odds scale which is not stated in the plot or the caption of Figure 3a nor in the caption of Table S2 which contains the effect sizes and credible intervals for the causal models, "causal salad" models, and models with no other covariates. Displaying these effect sizes as odds ratios rather than log-odds would make the results more easily interpretable. Furthermore, the figure caption states that the credible interval lines reflect the 95% highest posterior density intervals (HPDI), when in fact the code shows that these are rather 90% and 50% percentile intervals (PI). The caption of Table S2 also shows no indication of what the displayed credible intervals represent. The covariates included in the predictive model were the variables in the DAG which directly influenced the response. This follows from Pearl's Markov blanket approach (Pearl 1988).

2.5 Limitations of the study

Many of the limitations of the study have already been outlined above such as those regarding the timing of data collection as well as potentially excluding important variables from the DAG. A key question which remains unanswered is why there was a change in the causal model structure as illustrated by the inconsistencies between the DAG in the paper and the code. This change may have been a retrospective decision since some code and a few details in the supplementary information were not changed while others were. There is a possibility that the estimates from this causal model were more appealing and the results were cherry-picked or potentially during review of the paper new information regarding the relationships in the DAG were revealed leading to the change. Either way, the reasons for changes and explanations of the alternative DAGs should have been given.

There are also limitations when it comes to reproducibility as detailed in the *Reproduction* section below. These include the lack of a fully reproducible script and metadata for the supplied data, failing to set the seed when fitting the models, making assumptions for parameters such as for the use of 1000 samples and a threshold value of 0.5 for correct predictions for reefs during the posterior predictive check of the fitted prediction model. Further, the supplementary information does not concur with the detail in the paper in some places, such as in S1.3.

While minor, there are a few grammatical and typesetting issues in the paper such as errors in the beta numbering in the statistical model specifications in both the paper and supplementary text (for the branching coral model and initial macroalgae model) and a missing a heading for the branching coral model specification in S1.3.

3 Reproduction

In order to reproduce the analysis by Arif et al. (2022), I obtained the data and R code accompanying the publication from Figshare (Arif 2021) and modified the code in R version 4.6.1 (R Core Team 2026). As alluded to above, no metadata were supplied for the four datasets nor for the variables within them other than what could be drawn from the paper, the supplementary information, and the code. The four datasets comprised an unstandardised and a standardised version of the data related to the original 1998 bleaching event, an unstandardised dataset with 2014 data for prediction after the 2016 bleaching event, and the predictions from the Bayesian predictive model. There were no missing values in any of the datasets and I confirmed that the standardised dataset was derived as stated from the unstandardised version by subtracting the mean of each variable and dividing by two standard deviations.

3.1 Causal analysis

I did not reproduce the creation of the DAG as this requires extensive knowledge about coral reef systems. Additionally, the reasons for the directed relationships within the DAG are well outlined in the supplementary text. After reading in the DAG from dagitty.web using the *dagitty* package (Textor et al. 2016), I noted that the DAG in the code differed from that in the paper as previously mentioned. Thus, I modified the DAG layout code to reflect the DAG in the paper (Figure 1). I confirmed that there were 32 DAG-implied independencies, cross-checked them against those listed in the supplementary data, and confirmed that the observed data were consistent with the independencies ($p = 1$) after performing Holm-Bonferroni correction to p-values to account for multiple testing. To assess the selection of covariates in the models for each variable, I manually drew the DAG and used the backdoor criterion to determine which variables to adjust for. I also made use of the adjustmentSets function from *dagitty* to determine the minimal adjustment set for each of the variables (Textor et al. 2016). As noted above, the minimal adjustment set was used for all variables except the initial macroalgae model in the paper. For the remainder of the analysis, I made use of the models stated in the paper since these all still satisfied the backdoor criterion.

As in the analysis by Arif et al. (2022), I used the *rethinking* package (McElreath 2024) to fit Bayesian logistic regression models for each of the variables of interest with the response being 0 for recovering sites and 1 for regime shifting sites. I used the supplied standardised dataset which corresponded with standardisation applied to the original dataset. Weakly informative priors were used including a Cauchy(0,10) prior for the intercept terms and student_t(4,0,2.5) priors for slope coefficients as recommended by previous authors and utilised in the paper. I used three chains rather than a single chain as in Arif et al. (2022) to assess convergence. I also made use of the more recently introduced ulam tool from *rethinking* rather than map2stan for model fitting (McElreath 2024). I also used cmdstanr rather than rstan to run the chains which utilises Hamiltonian Monte Carlo (HMC) sampling. All chains converged with Rhat < 1.01 for all parameters and traceplots showed appropriate mixing.

I plotted the posterior median with 90% and 50% percentile intervals (PI) for the causal effect of each of the variables on regime shift trajectory on the logit scale (log-odds). The paper stated that the solid lines in the plot represented the 95% highest posterior density intervals (HPDI). However, the code and the actual plot indicated that PIs not HPDIs were calculated so this is what I replicated (Figure 2a). Marginal plots of the continuous predictor variables were plotted by refitting the causal Bayesian logistic regression models for each variable with the stan_glm function which uses faster quadratic approximation rather than Monte Carlo sampling and then computing the marginal means and 95% predictive intervals for the given predictor with ggpredict (Figure 2(b-h)). These figures were all reproducible as Figure 3 in Arif et al. (2022) although small differences may exist due to stochasticity in fitting the Bayesian models with MCMC and extracting samples from the quadratic approximations.

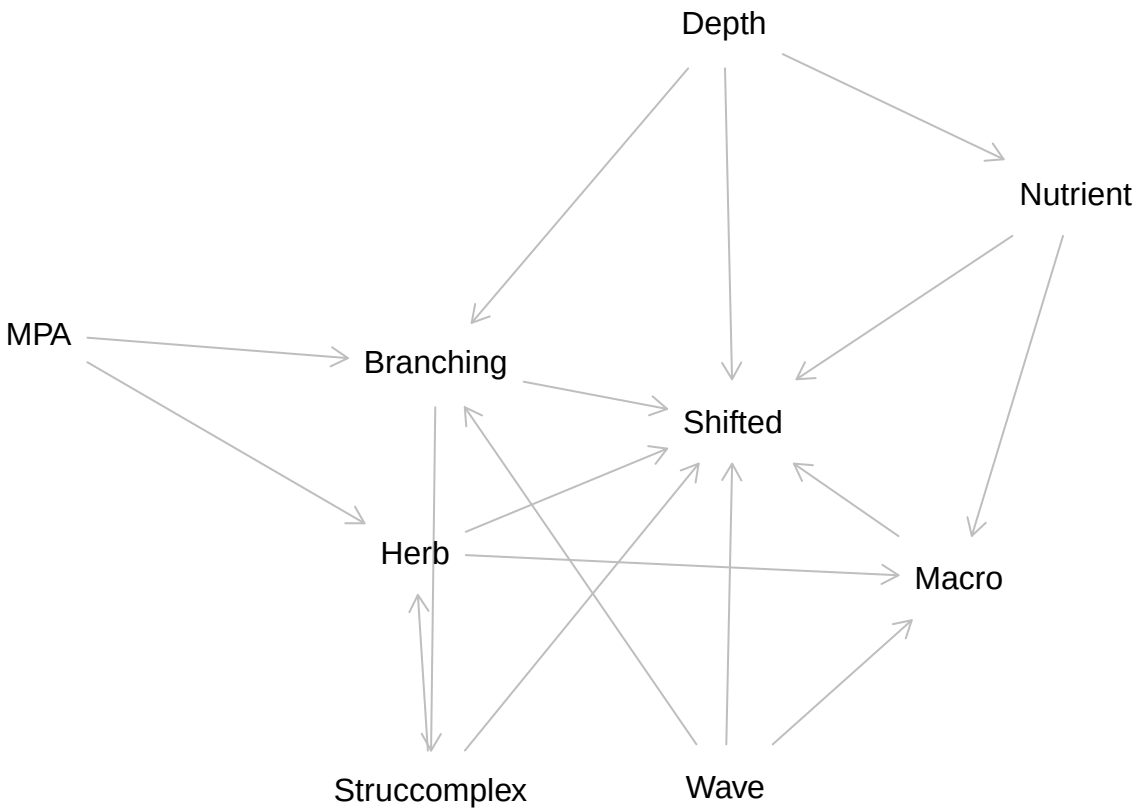


Figure 1: Directed acyclic graph (DAG) representing the causal structure of factors influencing regime shift versus recovery trajectories in Seychelles coral reefs. Herbivorous fish biomass, branching coral, macroalgal cover, and structural complexity represent predisturbance observational data (1994), whereas depth, MPA status, and wave exposure data represent values that are assumed to stay stable across years. There are some ambiguities regarding the year of collection of the nutrient data.

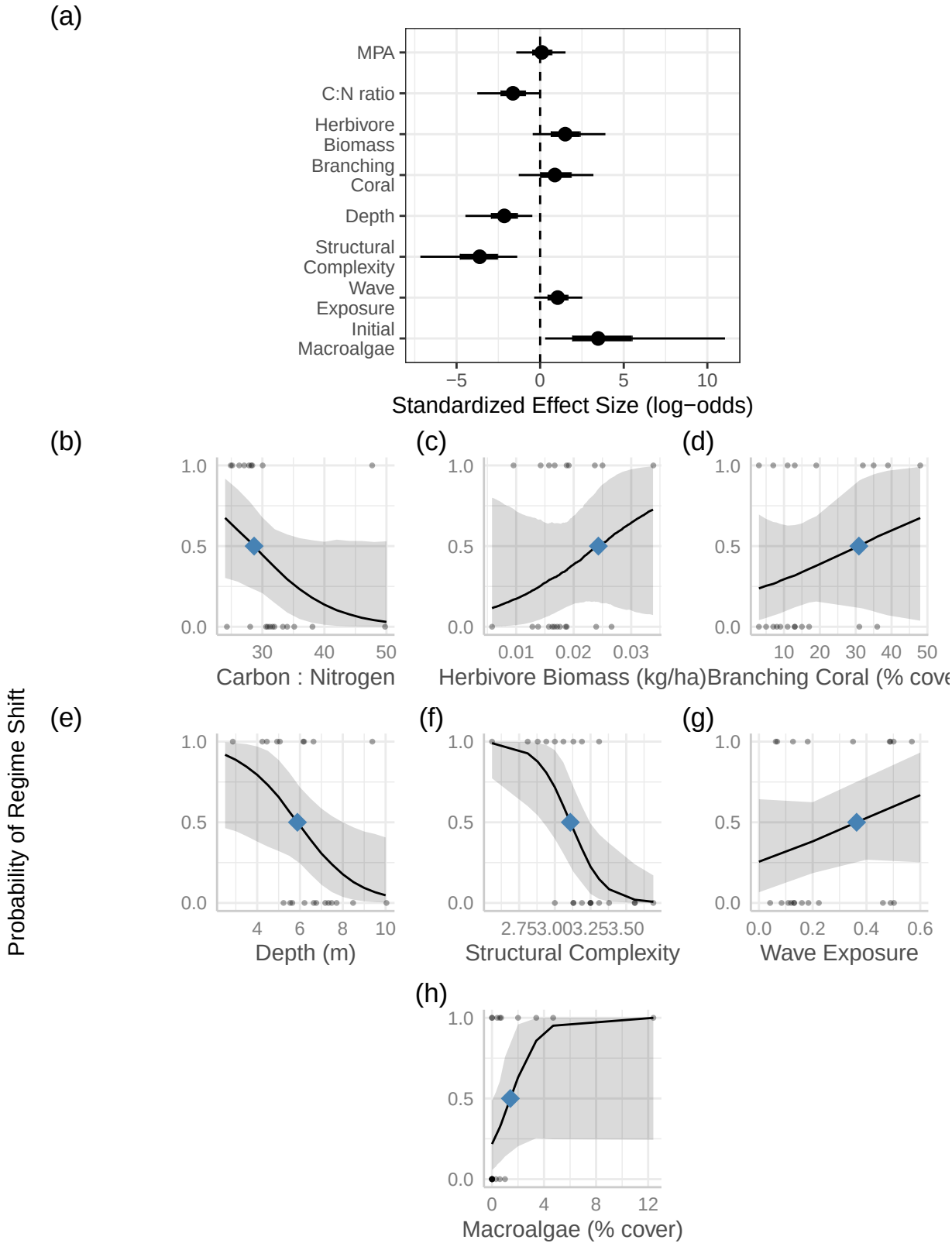


Figure 2: Causal effect of factors on regime shift trajectory. (a) Standardised effect size of factors influencing regime shift trajectory. Parameter estimates are posterior median values (dot) with 90% and 50% percentile intervals (PI) shown with thin line and thick lines respectively. (b-h) Marginal plots of predictor variables (for continuous variables) affecting regime shift trajectory: Solid black lines represent the predicted median of all drawn posterior predictive sampling; grey shading represents the 95% Bayesian predictive intervals; and blue dots represent the point at which regime shifts and recovery are equally likely.

3.2 DAG-informed predictive model

A causally guided predictive model was created in Arif et al. (2022) to determine the potential trajectory of 12 recovering reef sites after subsequent bleaching events in 2016. The predictor variables used by the authors comprised those directly influencing the response based on the DAG including depth, nutrients, branching coral, structural complexity, wave exposure, and initial macroalagal cover. Herbivorous fish biomass was excluded as a predictor since herbivorous fish biomass was not found to be a good proxy for herbivore grazing pressure. A Bayesian logistic regression model with weakly informative priors was used for the predictive model. The model was fitted to the unstandardised dataset from the 1998 bleaching event using the map function from *rethinking* (McElreath 2024). Normally distributed priors with mean 0 and standard deviation of 10 were used for both the intercept and slope coefficients. I verified that the model correctly predicted the recovery trajectory for 19 out of 21 sites as in Arif et al. (2022), using posterior predictive checks and assuming 1000 samples from the posterior and a threshold of 0.5. The recovery trajectories of Cousin Patch and St Anne Granite reefs were incorrectly predicted.

This trained model was used to predict recovery trajectories for the reefs which recovered post-1998 bleaching after the 2016 bleaching event using the 2014 dataset. The supplied code predicted on all 21 reefs using the link function with 1000 samples. I restricted the prediction to only the 12 recovering reefs and reproduced Figure 4a (Arif et al. 2022) by plotting the posterior probabilities (Figure 3a). I obtained similar results to the paper, with a few small differences in posterior means (for example, St Anne P and G) likely due to stochasticity in model fitting and random selection of the 1000 samples. I set the seed to ensure reproducibility of my analysis. Despite these small differences, these results confirmed that five reefs had a probability greater than 50% of shifting regime to macroalgae dominance based on their posterior means (Figure 3a). The marginal plots of the continuous predictor variables were replotted for the continuous variables included in the predictive model. For each variable, I overlaid data points from reefs where the variable's value corresponded to a regime-shift probability greater than 0.5. The ranges over which marginal means were predicted were extended for the structural complexity and macroalgae cover variables so as to include the overlaid data points (Figure 3(b-g)). The placement of the overlaid data points differed slightly to in the paper but once again this likely stems from stochasticity in the model fitting and with reproducing the exact figure being impossible as no seed was set.

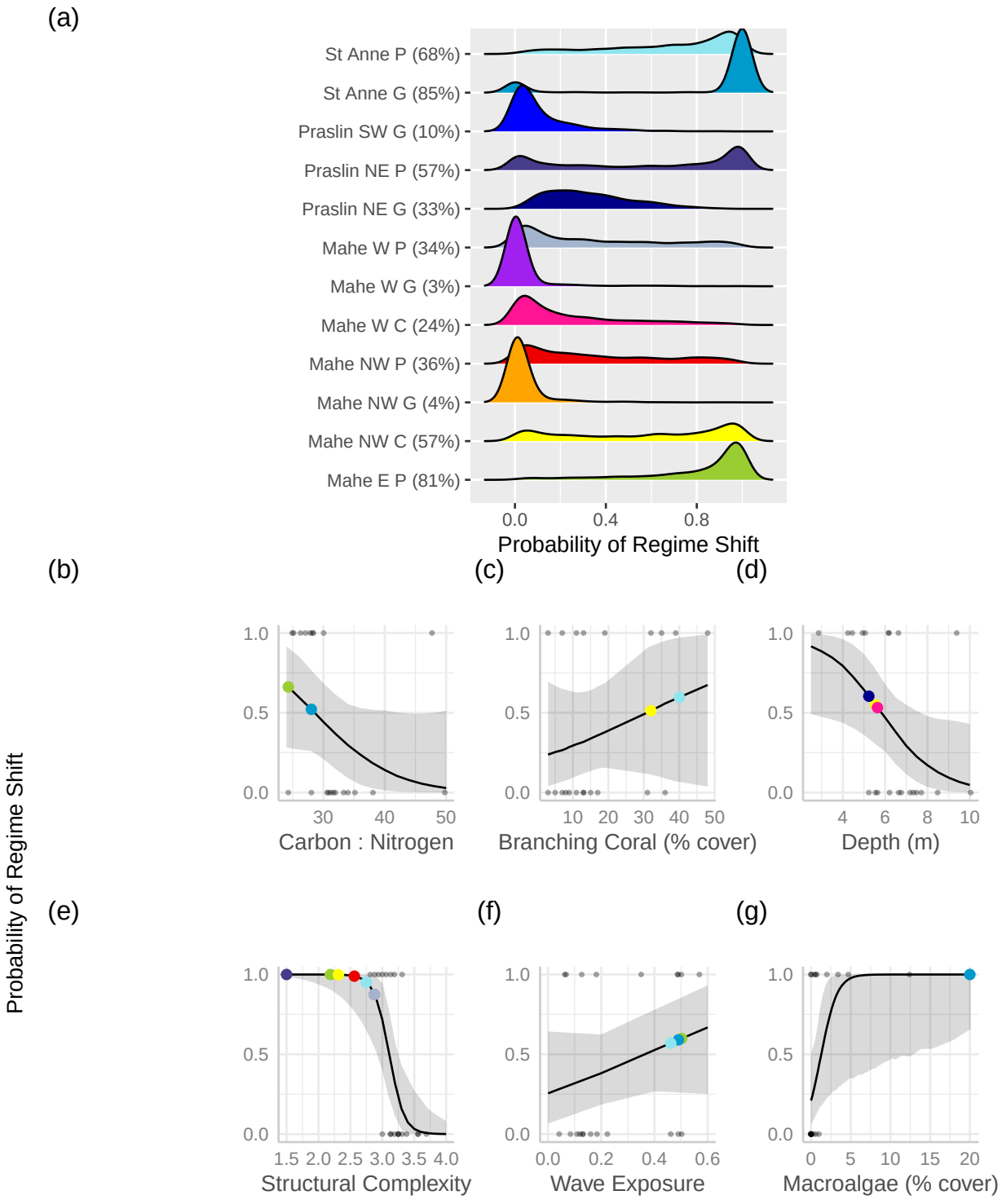


Figure 3: Predictions of future regime shifts in Seychelles. (a) Probability of regime shift for recovering sites across Seychelles after a subsequent bleaching event in 2016, based on our predictive model. Estimates are based on 1000 samples drawn from the posterior predictive distribution, with mean probability noted in brackets. (b–g) Marginal plots of predictor variables affecting regime shift trajectory, highlighting the 2014 predictor values for recovering reefs. Values that fall above the 0.5 probability line indicate risk factors for regime shifting after the 2016 bleaching event

Lastly, I reproduced Table S2 in the supplementary text which compared the causal model, "causal salad" model, and single variable model estimates (Table 1). This involved fitting a "causal salad" Bayesian logistic regression model using all the covariates and a Bayesian logistic regression model for each variable individually (for those variables that did not already have only a single variable in the causal model). As in Arif et al. (2022), the estimates varied across the different models, highlighting the importance of causal models. The estimates also differed slightly compared to Table S2 due to the stochasticity in the Bayesian models as previously discussed.

Table 1: Results from causal (backdoor criterion) vs correlative models (causal salad; no covariates), including posterior median values with 90% percentile intervals (PI) in brackets.

Predictor Variable	Backdoor Criterion	Causal Salad	No Covariates
MPA	0.10 [-1.43, 1.52]	-1.68 [-5.05, 0.95]	NA
Nutrient	-1.63 [-3.76, 0.02]	-1.63 [-2.70, 1.80]	-1.63 [-3.39, 0.28]
Herbivore Biomass	1.50 [-0.45, 3.90]	0.89 [-1.60, 4.18]	0.87 [-0.59, 2.47]
Branching Coral	0.88 [-1.28, 3.19]	-0.16 [-2.90, 2.40]	1.43 [-0.07, 3.18]
Depth	-2.14 [-4.47, -0.47]	-4.95 [-9.83, -1.28]	NA
Structural Complexity	-3.61 [-7.16, -1.37]	-4.83 [-9.71, -1.56]	-3.84 [-7.31, -1.54]
Wave Exposure	1.05 [-0.36, 2.52]	1.87 [-0.75, 5.36]	NA
Initial Macroalgae	3.47 [0.30, 11.06]	1.47 [-1.71, 6.37]	4.33 [1.20, 10.37]

4 Conclusion

The paper by Arif et al. (2022) correctly utilises well-developed methods in causal inference, including their use a DAG informed by domain knowledge, testing the d-separation conditions in their developed DAG, and employing the backdoor adjustment criterion appropriately. These all lend validity to their results. There are however, some instances where choices made in the causal inference analysis should be explained more clearly including the sources of data and the reason for evidence of alternative causal models. Having supplementary text, code, and data accompanying the paper is a step towards reproducible research. Nevertheless, there are inconsistencies and ambiguities in these sources which detract from the reproducibility and require some assumptions to replicate. Overall, the results need to be interpreted keeping these limitations of the study in mind.

References

Arif, Suchinta. 2021. "Causal Drivers of Climate-Mediated Coral Reef Regime Shifts." *Figshare*, ahead of print. https://doi.org/10.6084/m9.figshare.14981235.v1.

Arif, Suchinta, Nicholas A. J. Graham, Shaun Wilson, and M. Aaron MacNeil. 2022. "Causal Drivers of Climate-Mediated Coral Reef Regime Shifts." *Ecosphere* 13 (3): e3956. https://doi.org/10.1002/ecs2.3956.

Graham, Nicholas A. J., Simon Jennings, M. Aaron MacNeil, David Mouillot, and Shaun K. Wilson. 2015. "Predicting Climate-Driven Regime Shifts Versus Rebound Potential in Coral Reefs." *Nature* 518 (7537): 94–97. https://doi.org/10.1038/nature14140.

McElreath, Richard. 2020. *Statistical Rethinking: A Bayesian Course with Examples in r and Stan*. CRC Press.

McElreath, Richard. 2024. *Rethinking: Statistical Rethinking Book Package*. Version 2.42. https://github.com/rmcelreath/rethinking.

Pearl, Judea. 1988. *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference*. Morgan Kaufmann.

Pearl, Judea. 2009. "Causal Inference in Statistics: An Overview." *Statistics Surveys* 3: 96–146. https://doi.org/10.1214/09-SS057.

R Core Team. 2026. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. https://doi.org/10.32614/R.manuals.

Textor, Johannes, Benito van der Zander, Mark S Gilthorpe, Maciej Liśkiewicz, and George TH Ellison. 2016. "Robust Causal Inference Using Directed Acyclic Graphs: The r Package 'Dagitty'." *International Journal of Epidemiology* 45 (6): 1887–94. https://doi.org/10.1093/ije/dyw341.